USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LAB #9 (Lesson 28) - Null Steering Lab
Lab Sheet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 3.9: I can calculate null-steering weights and implement pattern nulls on the ADALM-PHASER.

This sheet is the turn-in document for the Lesson 28 lab. Every measurement below is a comparison against a frozen reference trace, so record the reference before you change anything. The lab is individual: one sheet per cadet, from your own kit.

1. **The static notch.** Compare the null-steered sweep against the frozen uniform reference. The levels at $+22.5^\circ$ are relative to their own trace's peak, and the main-lobe entries are relative to the uniform peak.

Quantity	Measured	Predicted
Reference level at $+22.5^\circ$		
Null-steered level at $+22.5^\circ$		
Change at that angle		
Main-lobe peak, uniform		
Main-lobe peak, null-steered		
Main-lobe cost		

2. **The difference null.** With Beam 1 Phase set to 180°, record the depth of the boresight null and the angles of the two peaks that replace the main lobe.

Quantity	Measured	Predicted
Null depth at boresight		
Left peak angle		
Right peak angle		

3. **MVDR against manual.** From your own procedure C sweeps, with the second HB100 at roughly $+30^\circ$ and running about 10 dB stronger than the boresight source. Report both levels relative to the peak of the manual trace.

Mode	Level at the interferer angle	Level in the look direction
Manual		
MVDR		
Difference		

4. **Written answers.** Three or four sentences each.

   (a) Why is the measured notch depth limited by the sweep's noise floor rather than by quantization? Give the numbers that set the floor, and the depth the quantized weights alone would allow.

   (b) Give one situation where a computed static null is the better choice than MVDR, and one where it is not, with a reason for each.

Documentation: _____